Sunflower protection built around existing tractors

Updated bands: 6 / 7 / 8 inches. Mechanical cultivation between rows; L&Aser; targets weeds in each crop-row band.

8.4 / 11.2 m
12 / 16 rows at 70 cm

21.8–29.0%
Field area assigned to laser at 70 cm rows

7 / 5
Calendar days / workable days per pass

Regional crop geometry: use crop type and local practice

Region / crop	Rows, cm	plants/ha	In-row, cm	Basis and source
Russia / oilseed	70	45,000–65,000	26.0	Harvest range; 55,000/ha reference. [R1,R2]
Russia / confectionery	70	30,000	47.6	Named varieties; 30,000/ha. [R1]
E. Europe / oilseed	70	50,000–60,000	26.0	50–60k reference; broader regional ranges. [E1–E3]
Serbia / confectionery	70	35,000–42,000	35.7	Harvest target; 40,000/ha model case. [E3,E4]
Heilongjiang / oilseed	70	37,500–45,000	35.7	Established stand; 70 cm row pitch is assumed. [C1]
Inner Mongolia / oilseed	60	55,556	30.0	Calculated 60×30 cm layout, not a reported stand. [C2]
Heilongjiang / confectionery	70	20,000	70.0	70×70 cm guidance; rounded stand target. [C1]

Seeds sown are distinct from harvest plants. For the Russian reference, 55,000 plants/ha plus 15–20% gives 63,250–66,000 seeds/ha. The source gives 25–30% on heavily weedy fields. Ukraine gives 5–10%; Bulgaria 10–15%. These are different regional recommendations. [R1,E1,E2]

Hungary also uses existing 75–76.2 cm rows. Xinjiang 40/90 cm paired rows under film and drip irrigation require a separate integration case; average pitch is not actual tool spacing. [E5,C3]

Match the planter, cultivator and laser band

Working swath = crop-row count × row spacing. Frame length and transport width are different measurements.

Crop rows	60 cm / m	70 cm / m	75 cm / m	Modules: 1 / 2 / 3 per row	Representative host evidence
6	3.60	4.20	4.50	6 / 12 / 18	KRN-4.2 [R2]
8	4.80	5.60	6.00	8 / 16 / 24	KRN-5.6; ALTAIR 5.6 [R2,E6]
12	7.20	8.40	9.00	12 / 24 / 36	KRN-8.4; Gaspardo HS-12 [R2,E7]
16	9.60	11.20	12.00	16 / 32 / 48	Gaspardo HS-16 [E7]

Host examples establish row-count families, not every listed spacing for every model. The KRN reference is 70 cm. Match the existing planter; a nominal 10 m toolbar is not automatically row-compatible. China has a documented six-row EC-Weeder demonstration, while its manufacturer offers custom widths up to 13 m. [R2,E7,E11,C4,C5]

Revised bands and the protected crop strip

Total laser band	Laser area at 70 cm rows	Against an 18 cm protected strip
152.4 mm	21.77%	Total uncovered gap: 27.6 mm
177.8 mm	25.40%	Total uncovered gap: 2.2 mm
203.2 mm	29.03%	Total overlap: 23.2 mm

The KRN manual specifies 9–16 cm clearance on EACH side. Thus 18 cm is the minimum total protected strip, not a universal setting. The 203.2 mm band covers 18–20 cm settings; wider clearances need a wider band or different tooling. Guided close-working tools provide a credible route for the narrower cases. [R3,E8]

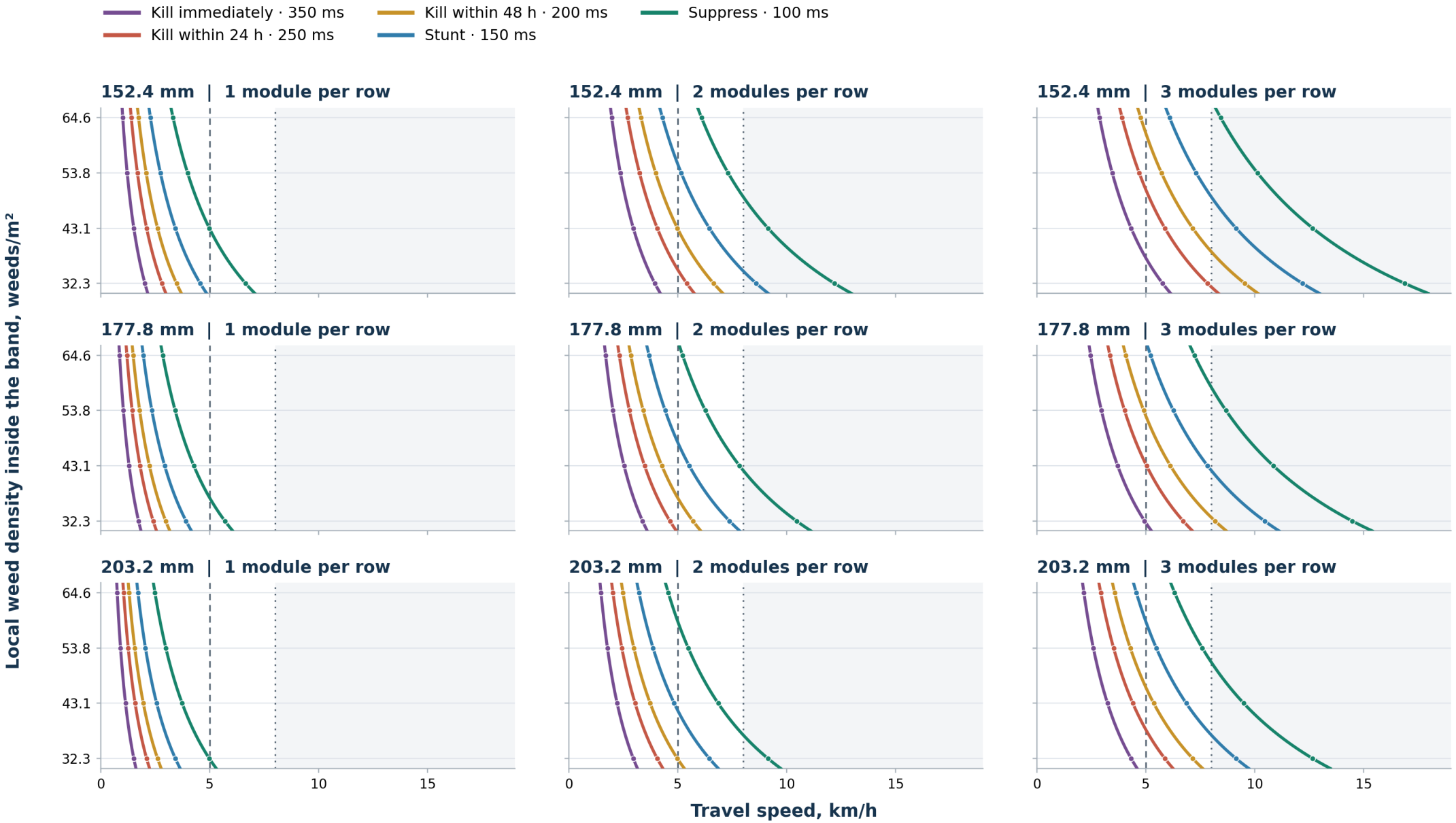
One-pass scheduling assumption

Base: 7 calendar days, including 5 workable days; 16 scheduled hours per workable day; 75% within-day efficiency. Stress/extended cases: 5 calendar / 3 workable days and 10 calendar / 7 workable days. These are planning assumptions, not regional weather forecasts or universal agronomic deadlines.

Cultivation speed: 3 km/h for delicate early stages; 5 km/h reference; 8 km/h conditional faster case. Stage-based guidance supports early passes and usually two, sometimes three interventions. A 30–40 day weed-sensitive phase is not the duration available for one pass. [R4,E9,E10,C1,C2]

L&Aser crop-row banding: laser capacity curves

Mechanical inter-row cultivation + 1–3 serial laser modules per crop row



Dashed: 5 km/h; dotted: 8 km/h. Cultivation planning speeds, separate from laser ceilings.

Dots mark all 180 requested cases. Per-stage cycle = total dwell / module count + 10 ms retarget time.

Curves assume ideal target handoffs. Outcome labels are user-supplied prescriptions; biological efficacy is not established by this model.

All 180 laser speed ceilings

Values in km/h. Read density group, then 1 / 2 / 3 serial modules per crop row.

Band	Total dwell, ms	32.29 weeds/m²			43.06 weeds/m²			53.82 weeds/m²			64.58 weeds/m²		
		1	2	3	1	2	3	1	2	3	1	2	3
	350	2.03	3.95	5.78	1.52	2.97	4.33	1.22	2.37	3.47	1.02	1.98	2.89
	250	2.81	5.42	7.84	2.11	4.06	5.88	1.69	3.25	4.70	1.41	2.71	3.92
152.4 mm	200	3.48	6.65	9.54	2.61	4.99	7.16	2.09	3.99	5.72	1.74	3.33	4.77
	150	4.57	8.61	12.19	3.43	6.45	9.14	2.74	5.16	7.32	2.29	4.30	6.10
	100	6.65	12.19	16.88	4.99	9.14	12.66	3.99	7.32	10.13	3.33	6.10	8.44
	350	1.74	3.39	4.95	1.31	2.54	3.71	1.05	2.03	2.97	0.87	1.69	2.48
	250	2.41	4.64	6.72	1.81	3.48	5.04	1.45	2.79	4.03	1.21	2.32	3.36
177.8 mm	200	2.99	5.70	8.18	2.24	4.28	6.13	1.79	3.42	4.91	1.49	2.85	4.09
	150	3.92	7.38	10.45	2.94	5.53	7.84	2.35	4.43	6.27	1.96	3.69	5.23
	100	5.70	10.45	14.47	4.28	7.84	10.85	3.42	6.27	8.68	2.85	5.23	7.23
	350	1.52	2.97	4.33	1.14	2.22	3.25	0.91	1.78	2.60	0.76	1.48	2.17
	250	2.11	4.06	5.88	1.58	3.05	4.41	1.27	2.44	3.53	1.06	2.03	2.94
203.2 mm	200	2.61	4.99	7.16	1.96	3.74	5.37	1.57	2.99	4.29	1.31	2.49	3.58
	150	3.43	6.45	9.14	2.57	4.84	6.86	2.06	3.87	5.49	1.71	3.23	4.57
	100	4.99	9.14	12.66	3.74	6.86	9.50	2.99	5.49	7.60	2.49	4.57	6.33

Blue cells support at least 5 km/h; green cells at least 8 km/h (3.11 / 4.97 mph). Counts: 58 of 180 reach 5; 20 reach 8. Actual field speed is the lower of this ceiling and the chosen cultivation cap.

350 ms: immediate kill; 250 ms: kill within 24 h; 200 ms: kill within 48 h; 150 ms: stunt; 100 ms: suppress. These are the requested dose/outcome hypotheses, not outcomes demonstrated by this calculation.

Capacity and fleet for 2,000 ha

Per pass; 75% within-day efficiency and 16 scheduled hours per workable day.

Reference: 203.2 mm, 53.82 weeds/m², 200 ms total, 3 modules per row

Rows / modules	Swath, m	Speed, km/h	ha/workday	One rig: workdays	Rigs: 3 / 5 / 7 workable days
8 / 24	5.60	4.29	28.9	69.3	24 / 14 / 10
12 / 36	8.40	4.29	43.3	46.2	16 / 10 / 7
16 / 48	11.20	4.29	57.7	34.7	12 / 7 / 5

The 12-row / 36-module reference needs 10 tractor-implement units for five workable days; the 16-row / 48-module version needs 7. These use the actual 4.29 km/h laser ceiling, not an assumed 5 or 8 km/h travel speed.

Higher-capacity prescriptions: a third stage can become redundant

Band / local density	Total dwell / stages	Laser ceiling, km/h	Chosen cap / field speed	ha/day: 12 rows	Rigs / 5 workdays
152.4 mm / 32.29/m²	100 ms / 1	6.65	5.00 / 5.00	50.4	8
152.4 mm / 32.29/m²	100 ms / 2	12.19	8.00 / 8.00	80.6	5
152.4 mm / 32.29/m²	100 ms / 3	16.88	8.00 / 8.00	80.6	5
203.2 mm / 32.29/m²	100 ms / 1	4.99	5.00 / 4.99	50.3	8
203.2 mm / 32.29/m²	100 ms / 2	9.14	8.00 / 8.00	80.6	5
203.2 mm / 32.29/m²	100 ms / 3	12.66	8.00 / 8.00	80.6	5

At the 203.2 mm low-density suppression case, two stages already support the faster cap. At that cap, 12 rows cover 80.6 ha/workday and 16 rows 107.5; fleets are 5 and 4 units respectively. More serial stages add capacity margin, not field area once cultivation is limiting.

Calculation method and design decisions

The model follows individual crop-row target demand and a balanced serial pipeline.

Per-row target stream
ρ = local weeds/m² inside the laser band; b = total band width in metres; T = total dwell per weed in seconds; n = serial modules per crop row; r = 0.010 s retarget time at every stage.
R = crop-row count; s = row spacing in metres; D = workable field days.

Per-stage cycle: $c = T/n + r$
Completed weeds per row: $q = 1/c$, weeds/s
Laser ceiling: $v = 3.6 / (\rho \times b \times c)$, km/h
Field swath: $W = R \times s$
Laser-treated area fraction: $f = b/s$
Field speed: min(v, cultivation speed cap)
Effective capacity: $A = W \times v_{field} / 10 \times 0.75$, ha/h
Per workable day: $A \times 16$
Fleet: round up $2{,}000 / (A \times 16 \times D)$.

Inherited workbook inputs
10 ms retarget time, 75% within-day efficiency and a 16-hour scheduled day. Crop-row target streams replace the earlier whole-width geometry. The old speed derating based on an implicit one-second interval is not retained as an established physical limit.

What a baton handoff means
Each weed receives total dwell T, not T from every stage. At 200 ms and three modules, each stage supplies 66.67 ms plus 10 ms retarget. Biological equivalence of separated exposures and a continuous dose remains to be tested.

Do not reduce density twice
Narrowing the band reduces targets per metre of travel; local density stays at 3 / 4 / 5 / 6 weeds/ft² (32.29 / 43.06 / 53.82 / 64.58 weeds/m²). Sunflower population affects crop avoidance and geometry, not this fixed weed-density input.

Recommended embodiment and practical boundaries

Use the existing planter-matched 8-, 12- or 16-row tractor host. Start the conventional KRN comparison at the 8-inch band, then select 1–3 stages for the actual prescription and density. For low-density suppression, spend additional capacity on useful swath before adding a redundant third stage. For longer dwell or denser weeds, stage capacity becomes the limiting factor.

These are mean-demand ceilings with ideal detection, tracking, dose sharing and handoff balance. Weed clusters, misses, crop occlusion, power/thermal limits and mechanical conditions can reduce performance. The 75% factor covers time losses; it does not prove on-the-move optical capacity. No extra efficiency factor is hidden in the curves.

Results are per pass over a synchronized crop cohort. Staggered sowing may spread demand; repeated weed flushes require additional passes. A 24/48-hour kill label is biological response time after exposure, not the time allowed to finish the field. Suppression may require more follow-up than a lethal treatment.

Primary sources and provenance

Research checked 7 November 2026. Click a title to open its source. Equipment offerings do not establish national fleet prevalence.

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